Victoria **Chui**

Ph.D. Candidate - University of Toronto
Human-Centered Data Science

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University of Toronto, Canada
ORCID Scholar
Website

Education

Ph.D. in Information **2023 – Present**
University of Toronto, Canada
Thesis: *Developing Human-Centered Algorithms to Assist Policy Decision Making in Peel Region Regarding Type 2 Diabetes*
Thesis supervisor: Dr. Shion Guha

Master of Information (MI) **2023**
University of Toronto, Canada
Thesis: *Towards Human-Centered Models in Healthcare: Development of Practical Applications for Surgical Wards*
Thesis supervisor: Dr. Shion Guha

Honours Bachelor of Science in Biochemistry (HBSc) **2021**
McGill University, Canada

Research Experience

Doctoral Researcher **2023 – Present**
Faculty of Information, University of Toronto
Supervisor: Dr. Shion Guha
- Analyzing Canadian health data to deconstruct and improve risk assessment models for diabetes care
- Understanding community trust/barriers regarding healthcare AI in Peel Region, Ontario

Research Assistant **2021 – 2023**
Faculty of Information, University of Toronto
Supervisor: Dr. Shion Guha
- Developed time series models to predict daily surgical case volumes
- Analyzed surgical site infection data using clustering and predictive methods
- Examined the landscape of Canadian child welfare resources and risk assessment models

Research Assistant **2020 – 2021**
Department of Biochemistry, McGill University
Supervisor: Dr. Bhushan Nagar
- Developed molecular dynamics simulations to examine IFIT protein and nonstandard-RNA interactions
- Executed simulations on GROMACS software, using Compute Canada ARC systems

Peer-Reviewed Publications

- **Victoria Chui**, Kelly McConvey, and Shion Guha. 2026. A Human-Centered Review of Algorithmic Design of AI in Health Systems and their Relevant Costs. In *GROUP2027*. [**Revise & Resubmit**]
- **Victoria Chui**, Kelly McConvey, Daniel Chui, Malayna Bernstein, and Shion Guha. 2026. Embedded Human-Centered Data Science in a Graduate Programming Course: A Framework and Case Study. *ACM Trans. Comput. Educ.* [**Revise & Resubmit**]
- **Victoria Chui**, Mara Silver, Mohammed Irfan, and Shion Guha. 2026. Teaching Human-Centered Data Science When There Is No Room in the Syllabus. In *Proceedings of the 58th ACM Technical Symposium on Computer Science Education V.2 (SIGCSE TS 2027), Vol. 2.* [**SUBMITTED**]
- **Victoria Chui**, Minahil Bakhtawar, Remziye Zaim, Ibukun-Oluwa Omolade Abejirinde, Jennifer L Gibson, Lorraine Lipscombe, Lori Diemert, Ijeoma Uchenna Itanyi, James Shaw, Laura C Rosella and Shion Guha. 2026. Auditing and Mitigating Bias in Population-Scale Diabetes Risk Prediction for Public Health Policy. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (October 2026).* [**SUBMITTED**]

- **Victoria Chui**, Sarah Rameez, Guanyu Mao, Edward Liu, Kelly Bang, Remziye Zaim, Ijeoma Uchenna Itanyi, Lori Diemert, Jennifer L Gibson, Lorraine Lipscombe, Ibukun-Oluwa Omolade Abejirinde, James Shaw, Laura C Rosella and Shion Guha. 2026. Advancing Responsible Public Health Policy through Human-Centered AI Design. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (October 2026).* [**SUBMITTED**]
- **Victoria Chui**, Kelly McConvey, Daniel Chui, Malayna Bernstein, and Shion Guha. 2026. Integrating Human-Centered Data Science into Computing Education: Insights from Semi-Structured Interviews. In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.2 (SIGCSE TS 2026), Vol. 2.* Association for Computing Machinery, New York, NY, USA, 1271–1272. doi.org/10.1145/3770761.3777231
- Ibukun-Oluwa Omolade Abejirinde, Ijeoma Uchenna Itanyi, Kathy Kornas, Remziye Zaim, Shion Guha, **Victoria Chui**, Lorraine Lipscombe, Laura C Rosella, and James Shaw. 2025. Principles and Practices of Community Engagement in AI for Population Health: Formative Qualitative Study of the AI for Diabetes Prediction and Prevention Project. *Journal of Participatory Medicine 17, (October 2025)*, e69497–e69497. doi.org/10.2196/69497
- Laura C Rosella, James Shaw, Shion Guha, Ibukun-Oluwa Omolade Abejirinde, Jennifer L Gibson, Lorraine Lipscombe, Kathy Kornas, Remziye Zaim, **Victoria Chui**, and Ijeoma Uchenna Itanyi. 2025. A participatory approach to deploy responsible artificial intelligence for diabetes prediction and prevention. *DIGITAL HEALTH 11, (2025)*, 20552076251358540. doi.org/10.1177/20552076251358541
- **Victoria Chui**, Kelly McConvey, Erina Seh-Young Moon, Maya Ghai, and Shion Guha. 2025. Towards Sustainable Community-Designed AI Systems in the Public Sector. In *ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS '25)*, July 22–25, 2025, Toronto, ON, Canada. ACM, New York, NY, USA, 837-840. doi.org/10.1145/3715335.3737683
- **Victoria Chui**, Jessica Pater, Tammy Toscos, and Shion Guha. 2023. Applying Human-Centered Data Science to Healthcare: Hyperlocal Modeling of COVID-19 Hospitalizations. In *Companion Proceedings of the 2023 ACM International Conference on Supporting Group Work (GROUP '23)*. Association for Computing Machinery, New York, NY, USA, 24–26. doi.org/10.1145/3565967.3570979
- **Victoria Chui** [Master's Thesis]. 2023. Towards Human-Centered Models in Healthcare: Development of Practical Applications for Surgical Wards. University of Toronto, Toronto, Canada.

Conference Presentations

- Abejirinde IO, Itanyi I, Kornas K, Zaim R, Guha S, **Chui V**, Lipscombe L, Rosella L, Shaw J (2025, May 25–29). Principles and Practices of Engaging Communities in Socially Responsible Deployment of Artificial Intelligence for Population Health: Findings from the AI for Diabetes Prediction and Prevention Project. *Canadian Association for Health Services and Policy Research Conference*, Ottawa, Canada.

Research Artifacts

Human-Centered Data Science Pedagogy Toolkit: Modules 1-4 **2025**
*Guha, S., **Chui, V.**, Silver, M., & Irfan, M.*
Impact: A code-based pedagogical toolkit for disseminating Human-Centered Data Science curricula. Currently being beta-tested by instructors at multiple universities to facilitate modular, interactive HCDS instruction. Represents a major contribution to Educational Leadership and SoTL

Teaching Experience

Course Instructor **University of Toronto**
INF1340: Programming for Data Science — Summer 2024, 2025, 2026
CCT111: Critical Coding – Summer 2026
INF2178: Experimental Design for Data Science – Winter 2026

Teaching Assistant **University of Toronto**
INF2178: Experimental Design for Data Science — Winter 2024, 2025
INF1339: Introduction to Computational Thinking – Fall 2023, 2024
CCT202: Human-Machine Communication – Fall 2023

Course Developer **Centennial College**

DATA702: Data Foundations — Fall 2024
DATA707: Qualitative Data – Fall 2024

Supervision

Direct PhD Supervisor **2025 – 2026**
University of Toronto, Canada
Student: Minahil Bakhtawar, Bachelor of Applied Science
Thesis: *Model Audit & Design: Human-Centered Approach to Developing Predictive Risk Models for Type 2 Diabetes*

Service

President – Doctoral Student Association, Faculty of Information 2024 – 2026
- Representing and advocating for doctoral student needs at the iSchool, University of Toronto

Faculty Hiring Search Committee Member, Faculty of Information 2022 – 2026
- Assisted search committees for full-time teaching-stream faculty members, student representative
- Collaborated on job posting descriptions, participated in shortlisting and interview rounds

Peer Reviewer
- Association of Computing Machinery (ACM)
  - Conference on Ethics, AI and Society (AIES)
  - Conference on Human Factors in Computing Systems (CHI)
  - Special Interest Group on Computer Science Education (SIGCSE)
  - Journal of Responsible Computing
- Information Processing & Management (IP&M) Journal

Volunteer Roles

Clinical Hospital Volunteer, Toronto Western Hospital 2023 – 2025
- Performed administrative duties and assisted patients at the Toronto Western Hospital Neurology Clinic

Conference Volunteer, International Conference on Dublin Core and Metadata Applications 2024

Speaking Engagements

Public Lecture **University of Toronto**
Health Informatics Seminar Series
Examining the Landscape of Human-Centered AI for Health Systems and Implications for Risk Prediction

Guest Lectures **University of Toronto**
WRI363 – Communicating in a World of Data, Univeristy of Toronto
Guest lecture on data visualization
INF3003 – Research in Information: Foundations and Design
Guest lecture on human-centered data science

Panel Guest **University of Toronto**
CV and Cover Letter Workshop for PhD Students, University of Toronto

Awards

Ontario Graduate Scholarship (Doctoral) – $15,000 2026-2027
Ontario Graduate Scholarship (Doctoral) – $15,000 2025-2026

PhD Enhancement Award, Faculty of Information – $1000 2024

Certificates

Course Design Certificate - *Teaching Assistants' Training Program, University of Toronto* 2024